

Interacting Multipoles of Rank 3 and 3:

$$T_{\alpha\beta\gamma\delta\epsilon\zeta}\Omega_{\alpha\beta\gamma}\Omega_{\delta\epsilon\zeta} = R_z^{-13} \cdot \frac{1}{225} \cdot \left(\begin{aligned} &+105 \cdot R_z^4 \cdot R_\beta \cdot R_\epsilon \cdot \delta(\alpha, \gamma) \cdot \delta(\delta, \zeta) \cdot \Omega_{\alpha\beta\gamma} \cdot \Omega_{\delta\epsilon\zeta} \\ &+105 \cdot R_z^4 \cdot R_\beta \cdot R_\epsilon \cdot \delta(\alpha, \delta) \cdot \delta(\gamma, \zeta) \cdot \Omega_{\alpha\beta\gamma} \cdot \Omega_{\delta\epsilon\zeta} \\ &+105 \cdot R_z^4 \cdot R_\beta \cdot R_\epsilon \cdot \delta(\alpha, \zeta) \cdot \delta(\gamma, \delta) \cdot \Omega_{\alpha\beta\gamma} \cdot \Omega_{\delta\epsilon\zeta} \\ &+105 \cdot R_z^4 \cdot R_\beta \cdot R_\zeta \cdot \delta(\alpha, \gamma) \cdot \delta(\delta, \epsilon) \cdot \Omega_{\alpha\beta\gamma} \cdot \Omega_{\delta\epsilon\zeta} \\ &+105 \cdot R_z^4 \cdot R_\beta \cdot R_\zeta \cdot \delta(\alpha, \delta) \cdot \delta(\gamma, \epsilon) \cdot \Omega_{\alpha\beta\gamma} \cdot \Omega_{\delta\epsilon\zeta} \\ &+105 \cdot R_z^4 \cdot R_\beta \cdot R_\zeta \cdot \delta(\alpha, \epsilon) \cdot \delta(\gamma, \delta) \cdot \Omega_{\alpha\beta\gamma} \cdot \Omega_{\delta\epsilon\zeta} \\ &+105 \cdot R_z^4 \cdot R_\gamma \cdot R_\delta \cdot \delta(\alpha, \beta) \cdot \delta(\epsilon, \zeta) \cdot \Omega_{\alpha\beta\gamma} \cdot \Omega_{\delta\epsilon\zeta} \\ &+105 \cdot R_z^4 \cdot R_\gamma \cdot R_\delta \cdot \delta(\alpha, \epsilon) \cdot \delta(\beta, \zeta) \cdot \Omega_{\alpha\beta\gamma} \cdot \Omega_{\delta\epsilon\zeta} \\ &+105 \cdot R_z^4 \cdot R_\gamma \cdot R_\delta \cdot \delta(\alpha, \zeta) \cdot \delta(\beta, \epsilon) \cdot \Omega_{\alpha\beta\gamma} \cdot \Omega_{\delta\epsilon\zeta} \\ &+105 \cdot R_z^4 \cdot R_\gamma \cdot R_\epsilon \cdot \delta(\alpha, \beta) \cdot \delta(\delta, \zeta) \cdot \Omega_{\alpha\beta\gamma} \cdot \Omega_{\delta\epsilon\zeta} \\ &+105 \cdot R_z^4 \cdot R_\gamma \cdot R_\epsilon \cdot \delta(\alpha, \delta) \cdot \delta(\beta, \zeta) \cdot \Omega_{\alpha\beta\gamma} \cdot \Omega_{\delta\epsilon\zeta} \\ &+105 \cdot R_z^4 \cdot R_\gamma \cdot R_\epsilon \cdot \delta(\alpha, \zeta) \cdot \delta(\beta, \delta) \cdot \Omega_{\alpha\beta\gamma} \cdot \Omega_{\delta\epsilon\zeta} \\ &+105 \cdot R_z^4 \cdot R_\gamma \cdot R_\zeta \cdot \delta(\alpha, \beta) \cdot \delta(\delta, \epsilon) \cdot \Omega_{\alpha\beta\gamma} \cdot \Omega_{\delta\epsilon\zeta} \\ &+105 \cdot R_z^4 \cdot R_\gamma \cdot R_\zeta \cdot \delta(\alpha, \delta) \cdot \delta(\beta, \epsilon) \cdot \Omega_{\alpha\beta\gamma} \cdot \Omega_{\delta\epsilon\zeta} \end{aligned} \right) \quad (1)$$

Simplify deltas:

$$T_{\alpha\beta\gamma\delta\epsilon\zeta}\Omega_{\alpha\beta\gamma}\Omega_{\delta\epsilon\zeta} = R_z^{-13} \cdot \frac{1}{225} \cdot \begin{aligned} & +105 \cdot R_z^4 \cdot R_\beta \cdot R_\epsilon \cdot \delta(\alpha, \alpha) \cdot \delta(\delta, \delta) \cdot \Omega_{\alpha\beta} \cdot \Omega_{\delta\delta\epsilon} \\ & +105 \cdot R_z^4 \cdot R_\beta \cdot R_\epsilon \cdot \delta(\alpha, \alpha) \cdot \delta(\gamma, \gamma) \cdot \Omega_{\alpha\beta\gamma} \cdot \Omega_{\alpha\gamma\epsilon} \\ & +105 \cdot R_z^4 \cdot R_\beta \cdot R_\epsilon \cdot \delta(\alpha, \alpha) \cdot \delta(\gamma, \gamma) \cdot \Omega_{\alpha\beta\gamma} \cdot \Omega_{\alpha\gamma\epsilon} \\ & +105 \cdot R_z^4 \cdot R_\beta \cdot R_\zeta \cdot \delta(\alpha, \alpha) \cdot \delta(\delta, \delta) \cdot \Omega_{\alpha\alpha\beta} \cdot \Omega_{\delta\delta\zeta} \\ & +105 \cdot R_z^4 \cdot R_\beta \cdot R_\zeta \cdot \delta(\alpha, \alpha) \cdot \delta(\gamma, \gamma) \cdot \Omega_{\alpha\beta\gamma} \cdot \Omega_{\alpha\gamma\zeta} \\ & +105 \cdot R_z^4 \cdot R_\beta \cdot R_\zeta \cdot \delta(\alpha, \alpha) \cdot \delta(\gamma, \gamma) \cdot \Omega_{\alpha\beta\gamma} \cdot \Omega_{\alpha\gamma\zeta} \\ & +105 \cdot R_z^4 \cdot R_\gamma \cdot R_\delta \cdot \delta(\alpha, \alpha) \cdot \delta(\epsilon, \epsilon) \cdot \Omega_{\alpha\alpha\gamma} \cdot \Omega_{\delta\epsilon\epsilon} \\ & +105 \cdot R_z^4 \cdot R_\gamma \cdot R_\delta \cdot \delta(\alpha, \alpha) \cdot \delta(\beta, \beta) \cdot \Omega_{\alpha\beta\gamma} \cdot \Omega_{\alpha\beta\delta} \\ & +105 \cdot R_z^4 \cdot R_\gamma \cdot R_\delta \cdot \delta(\alpha, \alpha) \cdot \delta(\beta, \beta) \cdot \Omega_{\alpha\beta\gamma} \cdot \Omega_{\alpha\beta\delta} \\ & +105 \cdot R_z^4 \cdot R_\gamma \cdot R_\epsilon \cdot \delta(\alpha, \alpha) \cdot \delta(\delta, \delta) \cdot \Omega_{\alpha\alpha\gamma} \cdot \Omega_{\delta\delta\epsilon} \\ & +105 \cdot R_z^4 \cdot R_\gamma \cdot R_\epsilon \cdot \delta(\alpha, \alpha) \cdot \delta(\beta, \beta) \cdot \Omega_{\alpha\beta\gamma} \cdot \Omega_{\alpha\beta\epsilon} \\ & +105 \cdot R_z^4 \cdot R_\gamma \cdot R_\epsilon \cdot \delta(\alpha, \alpha) \cdot \delta(\beta, \beta) \cdot \Omega_{\alpha\beta\gamma} \cdot \Omega_{\alpha\beta\epsilon} \\ & +105 \cdot R_z^4 \cdot R_\gamma \cdot R_\zeta \cdot \delta(\alpha, \alpha) \cdot \delta(\delta, \delta) \cdot \Omega_{\alpha\alpha\gamma} \cdot \Omega_{\delta\delta\zeta} \\ & +105 \cdot R_z^4 \cdot R_\gamma \cdot R_\zeta \cdot \delta(\alpha, \alpha) \cdot \delta(\beta, \beta) \cdot \Omega_{\alpha\beta\gamma} \cdot \Omega_{\alpha\beta\zeta} \end{aligned} \quad (2a)$$

$$\begin{aligned} & \xrightarrow{\text{added up}} R_z^{-13} \cdot \frac{1}{225} \cdot \\ & +210 \cdot R_z^4 \cdot R_\beta \cdot R_\gamma \cdot \Omega_{\alpha\beta\gamma} \cdot \Omega_{\alpha\delta\delta} \\ & +105 \cdot R_z^4 \cdot R_\beta \cdot R_\delta \cdot \Omega_{\alpha\alpha\beta} \cdot \Omega_{\delta\epsilon\epsilon} \\ & +210 \cdot R_z^4 \cdot R_\beta \cdot R_\delta \cdot \Omega_{\alpha\beta\gamma} \cdot \Omega_{\alpha\gamma\delta} \\ & +105 \cdot R_z^4 \cdot R_\beta \cdot R_\epsilon \cdot \Omega_{\alpha\alpha\beta} \cdot \Omega_{\delta\delta\epsilon} \\ & +210 \cdot R_z^4 \cdot R_\beta \cdot R_\epsilon \cdot \Omega_{\alpha\beta\gamma} \cdot \Omega_{\alpha\gamma\epsilon} \\ & +105 \cdot R_z^4 \cdot R_\beta \cdot R_\zeta \cdot \Omega_{\alpha\alpha\beta} \cdot \Omega_{\delta\delta\zeta} \\ & +210 \cdot R_z^4 \cdot R_\beta \cdot R_\zeta \cdot \Omega_{\alpha\beta\gamma} \cdot \Omega_{\alpha\gamma\zeta} \\ & +105 \cdot R_z^4 \cdot R_\gamma \cdot R_\delta \cdot \Omega_{\alpha\alpha\gamma} \cdot \Omega_{\delta\epsilon\epsilon} \\ & +210 \cdot R_z^4 \cdot R_\gamma \cdot R_\delta \cdot \Omega_{\alpha\beta\gamma} \cdot \Omega_{\alpha\beta\delta} \\ & +105 \cdot R_z^4 \cdot R_\gamma \cdot R_\epsilon \cdot \Omega_{\alpha\alpha\gamma} \cdot \Omega_{\delta\delta\epsilon} \\ & +210 \cdot R_z^4 \cdot R_\gamma \cdot R_\epsilon \cdot \Omega_{\alpha\beta\gamma} \cdot \Omega_{\alpha\beta\epsilon} \\ & +105 \cdot R_z^4 \cdot R_\gamma \cdot R_\zeta \cdot \Omega_{\alpha\alpha\gamma} \cdot \Omega_{\delta\delta\zeta} \\ & +210 \cdot R_z^4 \cdot R_\gamma \cdot R_\zeta \cdot \Omega_{\alpha\beta\gamma} \cdot \Omega_{\alpha\beta\zeta} \\ & +105 \cdot R_z^4 \cdot R_\delta \cdot R_\epsilon \cdot \Omega_{\alpha\alpha\gamma} \cdot \Omega_{\gamma\delta\epsilon} \\ & +105 \cdot R_z^4 \cdot R_\delta \cdot R_\epsilon \cdot \Omega_{\alpha\alpha\beta} \cdot \Omega_{\beta\delta\epsilon} \\ & +105 \cdot R_z^4 \cdot R_\delta \cdot R_\epsilon \cdot \Omega_{\alpha\beta\beta} \cdot \Omega_{\alpha\delta\epsilon} \\ & +105 \cdot R_z^4 \cdot R_\delta \cdot R_\zeta \cdot \Omega_{\alpha\alpha\gamma} \cdot \Omega_{\gamma\delta\zeta} \\ & +105 \cdot R_z^4 \cdot R_\delta \cdot R_\zeta \cdot \Omega_{\alpha\alpha\beta} \cdot \Omega_{\beta\delta\zeta} \\ & +105 \cdot R_z^4 \cdot R_\delta \cdot R_\zeta \cdot \Omega_{\alpha\beta\beta} \cdot \Omega_{\alpha\delta\zeta} \\ & +105 \cdot R_z^4 \cdot R_\epsilon \cdot R_\zeta \cdot \Omega_{\alpha\alpha\gamma} \cdot \Omega_{\gamma\epsilon\zeta} \\ & +105 \cdot R_z^4 \cdot R_\epsilon \cdot R_\zeta \cdot \Omega_{\alpha\alpha\beta} \cdot \Omega_{\beta\epsilon\zeta} \\ & +105 \cdot R_z^4 \cdot R_\epsilon \cdot R_\zeta \cdot \Omega_{\alpha\beta\beta} \cdot \Omega_{\alpha\epsilon\zeta} \\ & -15 \cdot R_z^6 \cdot \Omega_{\alpha\alpha\gamma} \cdot \Omega_{\gamma\epsilon\epsilon} \\ & 5-30 \cdot R_z^6 \cdot \Omega_{\alpha\alpha\gamma} \cdot \Omega_{\gamma\delta\delta} \\ & -15 \cdot R_z^6 \cdot \Omega_{\alpha\alpha\beta} \cdot \Omega_{\beta\epsilon\epsilon} \\ & -30 \cdot R_z^6 \cdot \Omega_{\alpha\alpha\beta} \cdot \Omega_{\beta\delta\delta} \end{aligned} \tag{2c}$$

Simplify R:

$$T_{\alpha\beta\gamma\delta\epsilon\zeta}\Omega_{\alpha\beta\gamma}\Omega_{\delta\epsilon\zeta} = R_z^{-13} \cdot \frac{1}{225}.$$

[illegible]

Reduce Indices:

$$T_{\alpha\beta\gamma\delta\epsilon\zeta}\Omega_{\alpha\beta\gamma}\Omega_{\delta\epsilon\zeta} = R_z^{-13} \cdot \frac{1}{225} \cdot \left[\begin{aligned} &-2835 \cdot R_z^6 \cdot \Omega_{zz\alpha} \cdot \Omega_{zz\alpha} - 2835 \cdot R_z^6 \cdot \Omega_{zz\alpha} \cdot \Omega_{zz\alpha} \\ &-2835 \cdot R_z^6 \cdot \Omega_{zz\alpha} \cdot \Omega_{zz\alpha} + 105 \cdot R_z^6 \cdot \Omega_{zz\alpha} \cdot \Omega_{\alpha\beta\beta} \\ &+ 210 \cdot R_z^6 \cdot \Omega_{zz\alpha} \cdot \Omega_{\alpha\beta\beta} + 105 \cdot R_z^6 \cdot \Omega_{zz\beta} \cdot \Omega_{\alpha\alpha\beta} \\ &+ 210 \cdot R_z^6 \cdot \Omega_{zz\beta} \cdot \Omega_{\alpha\alpha\beta} + 630 \cdot R_z^6 \cdot \Omega_{z\alpha\beta} \cdot \Omega_{z\alpha\beta} \\ &+ 105 \cdot R_z^6 \cdot \Omega_{zz\alpha} \cdot \Omega_{\alpha\beta\beta} + 210 \cdot R_z^6 \cdot \Omega_{zz\alpha} \cdot \Omega_{\alpha\beta\beta} \\ &+ 630 \cdot R_z^6 \cdot \Omega_{z\alpha\beta} \cdot \Omega_{z\alpha\beta} + 630 \cdot R_z^6 \cdot \Omega_{z\alpha\beta} \cdot \Omega_{z\alpha\beta} \\ &+ 315 \cdot R_z^6 \cdot \Omega_{zz\beta} \cdot \Omega_{\alpha\alpha\beta} + 315 \cdot R_z^6 \cdot \Omega_{zz\beta} \cdot \Omega_{\alpha\alpha\beta} \\ &+ 315 \cdot R_z^6 \cdot \Omega_{zz\alpha} \cdot \Omega_{\alpha\beta\beta} - 15 \cdot R_z^6 \cdot \Omega_{\alpha\alpha\gamma} \cdot \Omega_{\beta\beta\gamma} \\ &\quad - 30 \cdot R_z^6 \cdot \Omega_{\alpha\alpha\gamma} \cdot \Omega_{\beta\beta\gamma} \\ &\quad - 15 \cdot R_z^6 \cdot \Omega_{\alpha\alpha\beta} \cdot \Omega_{\beta\gamma\gamma} \\ &\quad - 30 \cdot R_z^6 \cdot \Omega_{\alpha\alpha\beta} \cdot \Omega_{\beta\gamma\gamma} \\ &\quad - 15 \cdot R_z^6 \cdot \Omega_{\alpha\beta\beta} \cdot \Omega_{\alpha\gamma\gamma} \\ &\quad - 90 \cdot R_z^6 \cdot \Omega_{\alpha\beta\gamma} \cdot \Omega_{\alpha\beta\gamma} \\ &\quad - 30 \cdot R_z^6 \cdot \Omega_{\alpha\beta\beta} \cdot \Omega_{\alpha\gamma\gamma} \end{aligned} \right] \quad (4a)$$

$$\xrightarrow{\text{added up}} R_z^{-13} \cdot \frac{1}{225} \cdot \left[\begin{aligned} &-8505 \cdot R_z^6 \cdot \Omega_{zz\alpha} \cdot \Omega_{zz\alpha} + 945 \cdot R_z^6 \cdot \Omega_{zz\alpha} \cdot \Omega_{\alpha\beta\beta} \\ &+ 945 \cdot R_z^6 \cdot \Omega_{zz\beta} \cdot \Omega_{\alpha\alpha\beta} + 1890 \cdot R_z^6 \cdot \Omega_{z\alpha\beta} \cdot \Omega_{z\alpha\beta} \\ &\quad - 45 \cdot R_z^6 \cdot \Omega_{\alpha\alpha\gamma} \cdot \Omega_{\beta\beta\gamma} \\ &\quad - 45 \cdot R_z^6 \cdot \Omega_{\alpha\alpha\beta} \cdot \Omega_{\beta\gamma\gamma} \\ &\quad - 45 \cdot R_z^6 \cdot \Omega_{\alpha\beta\beta} \cdot \Omega_{\alpha\gamma\gamma} \\ &\quad - 90 \cdot R_z^6 \cdot \Omega_{\alpha\beta\gamma} \cdot \Omega_{\alpha\beta\gamma} \end{aligned} \right] \quad (4b)$$

Multiply Out Sum (Duplikate Multipoles):

$$T_{\alpha\beta\gamma\delta\epsilon\zeta}\Omega_{\alpha\beta\gamma}\Omega_{\delta\epsilon\zeta} = R_z^{-13} \cdot \frac{1}{225} \cdot \left[\begin{aligned} &-8505 \cdot R_z^6 \cdot \Omega_{xzz} \cdot \Omega_{xzz} - 8505 \cdot R_z^6 \cdot \Omega_{yzz} \cdot \Omega_{yzz} \\ &-8505 \cdot R_z^6 \cdot \Omega_{zzz} \cdot \Omega_{zzz} + 945 \cdot R_z^6 \cdot \Omega_{zz\alpha} \cdot \Omega_{\alpha\beta\beta} \\ &+ 945 \cdot R_z^6 \cdot \Omega_{zz\beta} \cdot \Omega_{\alpha\alpha\beta} + 1890 \cdot R_z^6 \cdot \Omega_{xxz} \cdot \Omega_{xxz} \\ &+ 1890 \cdot R_z^6 \cdot \Omega_{xyz} \cdot \Omega_{xyz} + 1890 \cdot R_z^6 \cdot \Omega_{xzz} \cdot \Omega_{xzz} \\ &+ 1890 \cdot R_z^6 \cdot \Omega_{xyz} \cdot \Omega_{xyz} + 1890 \cdot R_z^6 \cdot \Omega_{yzz} \cdot \Omega_{yzz} \\ &+ 1890 \cdot R_z^6 \cdot \Omega_{yzz} \cdot \Omega_{yzz} + 1890 \cdot R_z^6 \cdot \Omega_{xzz} \cdot \Omega_{xzz} \\ &+ 1890 \cdot R_z^6 \cdot \Omega_{yzz} \cdot \Omega_{yzz} + 1890 \cdot R_z^6 \cdot \Omega_{zzz} \cdot \Omega_{zzz} \\ &\quad - 45 \cdot R_z^6 \cdot \Omega_{\alpha\alpha\gamma} \cdot \Omega_{\beta\beta\gamma} \\ &\quad - 45 \cdot R_z^6 \cdot \Omega_{\alpha\alpha\beta} \cdot \Omega_{\beta\gamma\gamma} \\ &\quad - 45 \cdot R_z^6 \cdot \Omega_{\alpha\beta\beta} \cdot \Omega_{\alpha\gamma\gamma} \\ &\quad - 90 \cdot R_z^6 \cdot \Omega_{xxx} \cdot \Omega_{xxx} - 90 \cdot R_z^6 \cdot \Omega_{xxy} \cdot \Omega_{xxy} \\ &\quad - 90 \cdot R_z^6 \cdot \Omega_{xxz} \cdot \Omega_{xxz} - 90 \cdot R_z^6 \cdot \Omega_{xxy} \cdot \Omega_{xxy} \\ &\quad - 90 \cdot R_z^6 \cdot \Omega_{xyy} \cdot \Omega_{xyy} - 90 \cdot R_z^6 \cdot \Omega_{xyz} \cdot \Omega_{xyz} \\ &\quad - 90 \cdot R_z^6 \cdot \Omega_{xxz} \cdot \Omega_{xxz} - 90 \cdot R_z^6 \cdot \Omega_{xyz} \cdot \Omega_{xyz} \\ &\quad - 90 \cdot R_z^6 \cdot \Omega_{xzz} \cdot \Omega_{xzz} - 90 \cdot R_z^6 \cdot \Omega_{xxy} \cdot \Omega_{xxy} \\ &\quad - 90 \cdot R_z^6 \cdot \Omega_{xyy} \cdot \Omega_{xyy} - 90 \cdot R_z^6 \cdot \Omega_{xyz} \cdot \Omega_{xyz} \\ &\quad - 90 \cdot R_z^6 \cdot \Omega_{xyy} \cdot \Omega_{xyy} - 90 \cdot R_z^6 \cdot \Omega_{yyy} \cdot \Omega_{yyy} \\ &\quad - 90 \cdot R_z^6 \cdot \Omega_{yyz} \cdot \Omega_{yyz} - 90 \cdot R_z^6 \cdot \Omega_{xyz} \cdot \Omega_{xyz} \\ &\quad - 90 \cdot R_z^6 \cdot \Omega_{yyz} \cdot \Omega_{yyz} - 90 \cdot R_z^6 \cdot \Omega_{yzz} \cdot \Omega_{yzz} \\ &\quad - 90 \cdot R_z^6 \cdot \Omega_{xxz} \cdot \Omega_{xxz} - 90 \cdot R_z^6 \cdot \Omega_{xyz} \cdot \Omega_{xyz} \\ &\quad - 90 \cdot R_z^6 \cdot \Omega_{xzz} \cdot \Omega_{xzz} - 90 \cdot R_z^6 \cdot \Omega_{xyz} \cdot \Omega_{xyz} \\ &\quad - 90 \cdot R_z^6 \cdot \Omega_{yyz} \cdot \Omega_{yyz} - 90 \cdot R_z^6 \cdot \Omega_{yzz} \cdot \Omega_{yzz} \\ &\quad - 90 \cdot R_z^6 \cdot \Omega_{xzz} \cdot \Omega_{xzz} - 90 \cdot R_z^6 \cdot \Omega_{yzz} \cdot \Omega_{yzz} \\ &\quad - 90 \cdot R_z^6 \cdot \Omega_{zzz} \cdot \Omega_{zzz} \end{aligned} \right] \quad (5a)$$

$$\xrightarrow{\text{cleaned up}} R_z^{-13} \cdot \frac{1}{225} \cdot \left[\begin{array}{l} -8505 \cdot R_z^6 \cdot \Omega_{yzz} \cdot \Omega_{yzz} + 945 \cdot R_z^6 \cdot \Omega_{zz\alpha} \cdot \Omega_{\alpha\beta\beta} \\ +945 \cdot R_z^6 \cdot \Omega_{zz\beta} \cdot \Omega_{\alpha\alpha\beta} + 1890 \cdot R_z^6 \cdot \Omega_{yzz} \cdot \Omega_{yzz} \\ +1890 \cdot R_z^6 \cdot \Omega_{yzz} \cdot \Omega_{yzz} - 45 \cdot R_z^6 \cdot \Omega_{\alpha\alpha\gamma} \cdot \Omega_{\beta\beta\gamma} \\ -45 \cdot R_z^6 \cdot \Omega_{\alpha\alpha\beta} \cdot \Omega_{\beta\gamma\gamma} \\ -45 \cdot R_z^6 \cdot \Omega_{\alpha\beta\beta} \cdot \Omega_{\alpha\gamma\gamma} \\ -90 \cdot R_z^6 \cdot \Omega_{xxy} \cdot \Omega_{xxy} - 90 \cdot R_z^6 \cdot \Omega_{xxy} \cdot \Omega_{xxy} \\ -90 \cdot R_z^6 \cdot \Omega_{xxy} \cdot \Omega_{xxy} - 90 \cdot R_z^6 \cdot \Omega_{yyy} \cdot \Omega_{yyy} \\ -90 \cdot R_z^6 \cdot \Omega_{yzz} \cdot \Omega_{yzz} - 90 \cdot R_z^6 \cdot \Omega_{yzz} \cdot \Omega_{yzz} \\ -90 \cdot R_z^6 \cdot \Omega_{yzz} \cdot \Omega_{yzz} \end{array} \right] \quad (5b)$$

$$\xrightarrow{\text{added up}} R_z^{-13} \cdot \frac{1}{225} \cdot \left[\begin{array}{l} -4995 \cdot R_z^6 \cdot \Omega_{yzz} \cdot \Omega_{yzz} + 945 \cdot R_z^6 \cdot \Omega_{zz\alpha} \cdot \Omega_{\alpha\beta\beta} \\ +945 \cdot R_z^6 \cdot \Omega_{zz\beta} \cdot \Omega_{\alpha\alpha\beta} - 45 \cdot R_z^6 \cdot \Omega_{\alpha\alpha\gamma} \cdot \Omega_{\beta\beta\gamma} \\ -45 \cdot R_z^6 \cdot \Omega_{\alpha\alpha\beta} \cdot \Omega_{\beta\gamma\gamma} \\ -45 \cdot R_z^6 \cdot \Omega_{\alpha\beta\beta} \cdot \Omega_{\alpha\gamma\gamma} \\ -270 \cdot R_z^6 \cdot \Omega_{xxy} \cdot \Omega_{xxy} - 90 \cdot R_z^6 \cdot \Omega_{yyy} \cdot \Omega_{yyy} \end{array} \right] \quad (5c)$$

Multiply Out Sum:

[illegible]

$$\begin{aligned}
& \xrightarrow{\text{cleaned up}} R_z^{-13} \cdot \frac{1}{225} \cdot \left[\begin{aligned}
& -4995 \cdot R_z^6 \cdot \Omega_{yzz} \cdot \Omega_{yzz} + 945 \cdot R_z^6 \cdot \Omega_{xxy} \cdot \Omega_{yzz} \\
& + 945 \cdot R_z^6 \cdot \Omega_{yyy} \cdot \Omega_{yzz} + 945 \cdot R_z^6 \cdot \Omega_{yzz} \cdot \Omega_{yzz} \\
& + 945 \cdot R_z^6 \cdot \Omega_{xxy} \cdot \Omega_{yzz} + 945 \cdot R_z^6 \cdot \Omega_{yyy} \cdot \Omega_{yzz} \\
& + 945 \cdot R_z^6 \cdot \Omega_{yzz} \cdot \Omega_{yzz} - 45 \cdot R_z^6 \cdot \Omega_{xxy} \cdot \Omega_{xxy} \\
& - 45 \cdot R_z^6 \cdot \Omega_{xxy} \cdot \Omega_{yyy} - 45 \cdot R_z^6 \cdot \Omega_{xxy} \cdot \Omega_{yzz} \\
& - 45 \cdot R_z^6 \cdot \Omega_{xxy} \cdot \Omega_{yyy} - 45 \cdot R_z^6 \cdot \Omega_{yyy} \cdot \Omega_{yyy} \\
& - 45 \cdot R_z^6 \cdot \Omega_{yyy} \cdot \Omega_{yzz} - 45 \cdot R_z^6 \cdot \Omega_{xxy} \cdot \Omega_{yzz} \\
& - 45 \cdot R_z^6 \cdot \Omega_{yyy} \cdot \Omega_{yzz} - 45 \cdot R_z^6 \cdot \Omega_{yzz} \cdot \Omega_{yzz} \\
& - 45 \cdot R_z^6 \cdot \Omega_{xxy} \cdot \Omega_{xxy} - 45 \cdot R_z^6 \cdot \Omega_{xxy} \cdot \Omega_{yyy} \\
& - 45 \cdot R_z^6 \cdot \Omega_{xxy} \cdot \Omega_{yzz} - 45 \cdot R_z^6 \cdot \Omega_{xxy} \cdot \Omega_{yyy} \\
& - 45 \cdot R_z^6 \cdot \Omega_{yyy} \cdot \Omega_{yyy} - 45 \cdot R_z^6 \cdot \Omega_{yyy} \cdot \Omega_{yzz} \\
& - 45 \cdot R_z^6 \cdot \Omega_{xxy} \cdot \Omega_{yzz} - 45 \cdot R_z^6 \cdot \Omega_{yyy} \cdot \Omega_{yzz} \\
& - 45 \cdot R_z^6 \cdot \Omega_{yzz} \cdot \Omega_{yzz} - 45 \cdot R_z^6 \cdot \Omega_{xxy} \cdot \Omega_{xxy} \\
& - 45 \cdot R_z^6 \cdot \Omega_{xxy} \cdot \Omega_{yyy} - 45 \cdot R_z^6 \cdot \Omega_{xxy} \cdot \Omega_{yzz} \\
& - 45 \cdot R_z^6 \cdot \Omega_{xxy} \cdot \Omega_{yyy} - 45 \cdot R_z^6 \cdot \Omega_{yyy} \cdot \Omega_{yyy} \\
& - 45 \cdot R_z^6 \cdot \Omega_{yyy} \cdot \Omega_{yzz} - 45 \cdot R_z^6 \cdot \Omega_{xxy} \cdot \Omega_{yzz} \\
& - 45 \cdot R_z^6 \cdot \Omega_{yyy} \cdot \Omega_{yzz} - 45 \cdot R_z^6 \cdot \Omega_{yzz} \cdot \Omega_{yzz} \\
& - 270 \cdot R_z^6 \cdot \Omega_{xxy} \cdot \Omega_{xxy} - 90 \cdot R_z^6 \cdot \Omega_{yyy} \cdot \Omega_{yyy}
\end{aligned} \right] \quad (6b)
\end{aligned}$$

$$\begin{aligned}
& \xrightarrow{\text{added up}} R_z^{-13} \cdot \frac{1}{225} \cdot \left[\begin{aligned}
& -3240 \cdot R_z^6 \cdot \Omega_{yzz} \cdot \Omega_{yzz} + 1620 \cdot R_z^6 \cdot \Omega_{xxy} \cdot \Omega_{yzz} \\
& + 1620 \cdot R_z^6 \cdot \Omega_{yyy} \cdot \Omega_{yzz} - 405 \cdot R_z^6 \cdot \Omega_{xxy} \cdot \Omega_{xxy} \\
& - 270 \cdot R_z^6 \cdot \Omega_{xxy} \cdot \Omega_{yyy} - 225 \cdot R_z^6 \cdot \Omega_{yyy} \cdot \Omega_{yyy}
\end{aligned} \right] \quad (6c)
\end{aligned}$$

Combining everything:

$$T_{\alpha\beta\gamma\delta\epsilon\zeta} \Omega_{\alpha\beta\gamma} \Omega_{\delta\epsilon\zeta} = \left[\begin{aligned}
& -\frac{72}{5} \cdot R_z^{-7} \cdot \Omega_{yzz} \cdot \Omega_{yzz} + \frac{36}{5} \cdot R_z^{-7} \cdot \Omega_{xxy} \cdot \Omega_{yzz} \\
& + \frac{36}{5} \cdot R_z^{-7} \cdot \Omega_{yyy} \cdot \Omega_{yzz} - \frac{9}{5} \cdot R_z^{-7} \cdot \Omega_{xxy} \cdot \Omega_{xxy} \\
& - \frac{6}{5} \cdot R_z^{-7} \cdot \Omega_{xxy} \cdot \Omega_{yyy} - 1 \cdot R_z^{-7} \cdot \Omega_{yyy} \cdot \Omega_{yyy}
\end{aligned} \right] \quad (7)$$